Supplementary Material

Wide-Scale Sampling with Hard-Argmin Selection and CBF Projection for Safe Quadrotor Navigation in Narrow Passages

1 Cross-Task Generalisation and Cross-Configuration Robustness

1.1 Cross-Task Generalisation

Table 1 reports cross-task results on TWO_GATE and U_SHAPE ($K = 10$, $N_{\mathrm{MC}} = 40$).

Table 1: Cross-task generalisation ($K = 10$, $N_{\mathrm{MC}} = 40$).

Task	Method	Succ.	TErr [m]	IAE
TWO_GATE	PD	40/40	0.045	5.84
TWO_GATE	$R_{1,\sigma=5}$	40/40	0.031	5.56
U_SHAPE	PD	40/40	0.054	6.82
U_SHAPE	$R_{1,\sigma=5}$	40/40	0.040	6.54

Bootstrap 95% CIs on ΔTErr and ΔIAE exclude zero for both tasks ($\Delta\mathrm{TErr} \approx -0.015$ [-0.022, -0.008], $\Delta\mathrm{IAE} \approx -0.29$ [-0.37, -0.21]).

1.2 Cross-Configuration Robustness

Table 2 provides full cross-configuration results with bootstrap CIs. Controllers use $K = 10$, $N_{\mathrm{MC}} = 40$, and are not re-tuned. The PD nominal uses $m = 1.5\,\mathrm{kg}$ throughout.

Table 2: Cross-configuration robustness on NARROW ($K = 10$, $N_{\mathrm{MC}} = 40$). $\Delta = R_{1,\sigma=5} - \mathrm{PD}$ on paired both-success trials.

Config.	PD succ.	$R_{1,\sigma=5}$ succ.	McN. (b, c)	ΔTErr [95% CI]	ΔIAE [95%
nominal $(1.5, 0.1)$	27/40	**40/40**	$(\mathbf{13}, \mathbf{0})$	$\mathbf{-0.094}$ [$\mathbf{-0.125}$, $\mathbf{-0.065}$]	$\mathbf{-2.02}$ [$\mathbf{-2.23}$,
+50% mass $(2.25, 0.1)$	0/40	31/40	$(\mathbf{31}, \mathbf{0})$	—	—
+50% drag $(1.5, 0.15)$	26/40	38/40	$(\mathbf{12}, \mathbf{0})$	$\mathbf{-0.093}$ [$\mathbf{-0.124}$, $\mathbf{-0.062}$]	$\mathbf{-1.91}$ [$\mathbf{-2.19}$,

McNemar $p < 0.001$ (bold). "—" $= n_{\mathrm{both}} = 0$. Under +50% mass, every PD failure was an $R_{1,\sigma=5}$ success ($b = 31$, $c = 0$).

1.3 Additional Baselines and Three-Seed Replication

Table 3 reports PI-PD, oracle Adaptive-PD, and single-source-wide variants under +50% mass.

Three-seed replication for +50% mass: pooled $R_{1,\sigma=5}$ $99/120 = 82.5\%$ (Wilson 95% CI $[74.9\%, 88.2\%]$), PD 0/120.

Table 3: Additional baselines on NARROW, +50% mass ($K = 10$, $N_{\mathrm{MC}} = 40$).

Method	Succ.	Notes
PD ($\sigma = 2$, nominal mass)	0/40	baseline
PI-PD ($K_i = 2$)	0/40	integral correction insufficient
Adaptive-PD (oracle mass)	28/40	requires oracle knowledge
$R_{1,\sigma=5}$ (two-source)	31/40	matches oracle ($p = 0.45$)
$R_{0,\sigma=5}$ (single-source-wide)	32/40	equivalent to $R_{1,\sigma=5}$ ($p = 0.508$)

1.4 Main Narrow-Passage Result: Full Table with CIs

Table **??** reproduces the complete main result including Wilson CIs, McNemar discordant counts, and bootstrap CIs on ΔTErr and ΔIAE.

Table 4: Main result on NARROW ($N_{\mathrm{MC}} = 80$ paired trials, full). Bold = significant ($p < 0.05$, CI excludes zero).

K	Method	succ. (Wilson 95% CI)	TErr	IAE	McNemar (b, c)	ΔTErr [95% CI]	ΔIAE [95% CI]
5	PD	$46/80 = 57.5\%$ $[46.4, 67.9]$	0.644	9.31	—	—	—
5	$R_{1,\sigma=5}$	$70/80 = 87.5\%$ $[78.5, 93.0]$	0.171	6.94	$(3, 27), p{<}0.001$	-3.04 $[-4.26, -1.92]$	-5.11 $[-6.51, -3.7$
10	PD	$53/80 = 66.3\%$ $[55.4, 75.6]$	0.380	8.53	—	—	—
10	$R_{1,\sigma=5}$	$79/80 = 98.8\%$ $[93.2, 99.8]$	0.039	6.22	$(1, 27), p{<}0.001$	-3.29 $[-4.40, -2.29]$	-5.64 $[-6.95, -4.3$
20	PD	$59/80 = 73.8\%$ $[63.1, 82.2]$	0.273	8.03	—	—	—
20	$R_{1,\sigma=5}$	$77/80 = 96.3\%$ $[89.6, 98.7]$	0.040	5.82	$(1, 19), p{<}0.001$	-2.29 $[-3.28, -1.30]$	-4.54 $[-5.75, -3.3$

1.5 TSH-NMPC Algorithm and Default Hyperparameters

Algorithm 1.4 summarises one closed-loop step of TSH-NMPC. Table 4 lists the default hyperparameters used throughout Section 5 of the main text.

[h] [1] state x_t, setpoint x_{sp}, budget K, scale σ_r, horizon S $u_{\mathrm{pd}} \leftarrow m(K_p(p_{\mathrm{sp}} - p_t) + K_d(v_{\mathrm{sp}} - v_t))$ PD nominal sample $u^{(1:K)} \sim \mathcal{N}(u_{\mathrm{pd}}, \sigma_r^2 I)$ wide-scale PD-centred clip $u^{(1:K)}$ to $[u_{\min}, u_{\max}]$ $k = 1, \dots, K$ **(vectorized)** simulate S-step rollout; compute $J^{(k)}$ via cost $u_{\mathrm{nom}} \leftarrow u^{(\arg\min_k J^{(k)})}$ hard-argmin build $(A(x_t), b(x_t))$ from DT-CCBF $u_{\mathrm{safe}} \leftarrow$ solve QP; fallback if infeasible u_{safe} Two-source variant: replace Line 3 with $K_A \leftarrow \lfloor K/2 \rfloor$ tight samples at $\sigma_{\mathrm{pd}} = 2$ plus $K_B = K - K_A$ wide samples at σ_r.

Table 5: TSH-NMPC hyperparameters (defaults used throughout the experiments).

Symbol	Default	Role
K	10	total candidate budget
σ_r	5.0 N	wide-scale noise (per axis; operating plateau $[5, 8]$)
σ_{pd}	2.0 N	tight-scale noise (legacy; two-source variant only)
S	20 (= 0.4 s)	rollout horizon
κ	5	DT-CCBF LSE smoothness

1.6 σ_r Sweep: Full Table

Table 5 provides the complete σ_r sweep with bootstrap CIs on ΔTErr and ΔIAE vs. PD.

Table 6: σ_r sweep on NARROW ($K = 10$, $N_{\mathrm{MC}} = 40$). PD baseline: 27/40, TErr = 0.294, IAE = 8.52.

σ_r [N]	succ.	TErr [m] (succ.)	IAE (succ.)	McNemar (b, c), p vs PD	ΔTErr, Δ
1	$30/40 = 75.0\%$	0.261	8.00	$(5, 2)$, 0.453 (n.s.)	-0.043 $[-0.070, -0.$
2	$36/40 = 90.0\%$	0.236	7.54	$(9, 0)$, **0.004**	-0.065 $[-0.092, -0.$
3	$34/40 = 85.0\%$	0.195	7.28	$(9, 2)$, 0.065 (marg.)	-0.065 $[-0.094, -0.$
5	$\mathbf{40/40 = 100\%}$	**0.036**	**6.23**	$(13, 0)$, $<$ **0.001**	$\mathbf{-0.094}$ $[\mathbf{-0.125}, \mathbf{-0.0}$
8	$\mathbf{40/40 = 100\%}$	**0.016**	**5.33**	$(13, 0)$, $<$ **0.001**	$\mathbf{-0.094}$ $[\mathbf{-0.126}, \mathbf{-0.0}$

2 MPPI, CEM, and iCEM Sweep Data

Table 6 compares MPPI, CEM, iCEM, and hard-argmin at $K = 10$, $\sigma \in \{2, 5\}$. Table 7 reports the MPPI temperature sweep.

Table 7: MPPI/CEM/iCEM vs. hard-argmin on NARROW ($N_{\mathrm{MC}} = 40$, seed 7777).

Method	σ	n_{iter}	Succ. (nominal)	Succ. (+50% mass)	TErr (nominal) [m]
MPPI ($\lambda = 0.1$)	2	1	22/40	1/40	~ 0.25
MPPI ($\lambda = 5.0$)	5	1	39/40	—	~ 0.16
CEM	2	3	24/40	2/40	~ 0.20
CEM	5	3	**40/40**	—	0.110
iCEM	2	3	**40/40**	—	~ 0.043
$R_{1,\sigma=5}$ (hard-argmin)	5	1	**40/40**	**32/40**	**0.036**

iCEM latency ≈ 58.8 ms vs. ≈ 6 ms for hard-argmin. CEM uses $3\times$ the effective rollout budget at $n_{\mathrm{iter}} = 3$.

Table 8: MPPI λ sweep on NARROW ($K = 10$, $\sigma = 2$, $N_{\mathrm{MC}} = 80$, seed 7777).

λ	Succ.	TErr [m]	IAE
0.01	$41/80 = 51.3\%$	0.455	8.48
0.05	$44/80 = 55.0\%$	0.424	8.36
0.1	$46/80 = 57.5\%$	0.392	8.22
0.5	$41/80 = 51.3\%$	0.442	8.52
1.0	$45/80 = 56.3\%$	0.425	8.41
5.0	$44/80 = 55.0\%$	0.480	8.53

The narrow-scale PD baseline at $K = 10$, $\sigma = 2$ achieves 66.3% (Table 2, main text). At matched $\sigma = 2$, MPPI does not reach the PD baseline.

3 Negative Ablation: Learned Prior vs. Random Wide Source

3.1 $N_{\mathrm{MC}} = 80$ Full Table

Table 8 provides the full $N_{\mathrm{MC}} = 80$ head-to-head (seed 7777). A_3 uses a 27,935-parameter TRM as Source B; the random source uses $\sigma = 5$.

Table 9: Negative ablation on NARROW ($N_{\mathrm{MC}} = 80$, seed 7777).

K	Method	Succ. / 80	TErr [m]	IAE	vs. $R_{1,\sigma=5}$ p
5	PD	$34/80 = 42.5\%$	0.117	8.29	—
5	A_3	$63/80 = 78.8\%$	0.069	6.38	0.572 (n.s.)
5	$R_{1,\sigma=5}$	$67/80 = 83.8\%$	0.044	6.65	—
10	PD	$54/80 = 67.5\%$	0.110	7.91	—
10	A_3	$73/80 = 91.2\%$	0.030	5.92	0.180 (n.s.)
10	$R_{1,\sigma=5}$	$78/80 = 97.5\%$	0.034	6.23	—
20	PD	$70/80 = 87.5\%$	0.084	7.52	—
20	A_3	$75/80 = 93.8\%$	0.022	5.65	0.219 (n.s.)
20	$R_{1,\sigma=5}$	$79/80 = 98.8\%$	0.016	5.75	—

A_3 vs. $R_{1,\sigma=5}$ is underpowered at this sample size ($\Delta_{\min} \approx 0.12$; observed $p \geq 0.180$).

3.2 $N_{\mathrm{MC}} = 300$ Headline

$N_{\mathrm{MC}} = 300$ (seed 7777): $R_{1,\sigma=5}$ $293/300 = 97.7\%$; $R_{0,\sigma=5}$ $296/300 = 98.7\%$; A_3 $266/300 = 88.7\%$. McNemar A_3 vs. $R_{1,\sigma=5}$: $p = 7.4 \times 10^{-6}$ (minimum detectable $\Delta_{\min} \approx 0.06$).

3.3 TRM Architecture

The TRM is a weight-shared latent recursive network (64-dim latent, 16 recursive steps, 27,935 parameters) mapping $[x_t, x_{\mathrm{sp}}]$ to a 30-dim control sequence. Only the first 3 dims are used for candidate generation. Training: 50,000 samples from 500 closed-loop PD+CBF trajectories. A second TRM trained on NARROW trajectories (9× better single-step TErr) yielded worse closed-loop A_3, consistent with diversity collapse of a sharper proposal.

4 CasADi+IPOPT NLP NMPC Details

4.1 NLP Formulation

Multiple-shooting with CasADi Opti + IPOPT, using identical Q, R weights and soft obstacle penalty ($\rho = 2000$, $\delta = 0.3\,\mathrm{m}$) as TSH-NMPC (Eq. 6, main text). Horizons: $H = 10$ (0.2 s) and $H = 20$ (0.4 s). Unlike TSH-NMPC, no separate CBF projection; the soft obstacle penalty is part of the NLP cost. Latency includes full Opti stack overhead and IPOPT solve on the same single-threaded x86 CPU.

4.2 Full Latency Distribution

Table 9 reports per-step latency from a profiling run ($N_{\mathrm{MC}} = 40$, 150 steps each).

4.3 Full Comparison Table

Table 10 reproduces the complete CasADi comparison including H=10, P99, and deadline-overrun columns omitted from the compact version in the main text.

Table 10: Per-step latency distribution on NARROW ($N_{\mathrm{MC}} = 40$, 150 steps).

Metric	CasADi H=10	CasADi H=20	$R_{1,\sigma=5}$ K=10	PD K=10
Mean [ms]	3.20	5.16	10.72	19.61
Median [ms]	2.66	3.74	10.67	17.28
Std [ms]	1.27	2.75	2.13	8.38
CV	0.40	0.53	0.20	0.43
p95 [ms]	5.73	10.17	14.21	34.69
p99 [ms]	6.89	11.91	16.25	54.25
Max [ms]	19.96	35.02	32.35	152.71
% > 20 ms	0.00%	0.12%	0.17%	31.60%

Table 11: TSH-NMPC vs. CasADi+IPOPT NMPC (full). "P99" = per-step 99th-percentile latency [ms]; "%>20" = fraction exceeding 20 ms. McNemar (b, c, p) vs. CasADi-$H20$.

Method	Succ.	TErr	Lat	P99	%>20	McN. vs. $H20$
Nominal NARROW $N_{\mathrm{MC}} = 80$*:*						
CasADi-$H10$	1/80	0.746	3.4	6.9	0.00	—
CasADi-$H20$	**80/80**	**0.003**	5.3	11.9	0.12	(ref.)
$R_{1,\sigma=5}$ K=10	77/80	0.063	5.7	—	—	$(3, 0)$, $p = 0.25$
$R_{1,\sigma=5}$ K=20	80/80	0.032	~ 6	—	—	$(0, 0)$, $p = 1.0$
$R_{0,\sigma=5}$ K=50	80/80	0.015	~ 10	—	—	$(0, 0)$, $p = 1.0$
+50% *mass* $N_{\mathrm{MC}} = 40$*:*						
CasADi-$H20$	37/40	0.054	5.3	11.9	0.12	(ref.)
$R_{0,\sigma=5}$ K=50	38/40	0.079	~ 10	—	—	$(1, 2)$, $p = 1.0$
$R_{0,\sigma=5}$ K=150	**40/40**	**0.044**	~ 20	—	—	$(0, 3)$, $p = 0.25$

Latency: per-trial mean-of-per-step-means from the $N_{\mathrm{MC}} = 80$ run; the profiling run (Table 9) yields per-step means of 10.7 ms (K=10), ~12 ms (K=20), ~20 ms (K=50).

5 Robustness, CBF Fallback, and Multiple Comparisons

5.1 CBF Fallback Audit

Table 11 reports CBF QP infeasibility rates on NARROW ($N_{\mathrm{MC}} = 40$). Fallback triggers the box-constrained emergency backup ($\delta_{\mathrm{buffer}} = 0.15$ m).

5.2 Dryden Wind Gust

MIL-F-8785C turbulence spectrum: base RMS $[0.5, 0.5, 0.3]$ m/s, length scales $L = [200, 200, 50]$ m, airspeed 5.0 m/s. Intensity 1×–5×. First-order gust filters apply $F_d = -b \cdot v_{\mathrm{gust}}$ ($b = 0.1$). On NARROW ($N_{\mathrm{MC}} = 40$, $K = 10$): both wide-σ controllers retain 39–40/40, PD drops to 26–28/40 ($p \leq 9.8 \times 10^{-4}$).

5.3 Dynamic Obstacle

Centre obstacle (index 1) moves at $v_y \leq 1.0$ m/s without predictive knowledge. Wide-σ controllers retain 39–40/40, matching CasADi+IPOPT $H = 20$ ($p = 1.0$) at $N_{\mathrm{MC}} = 40$, $K = 10$.

Table 12: CBF fallback invocations on NARROW ($N_{\text{MC}} = 40$).

Config.	Method	K	Succ.	Fallback rate
nominal	PD	5	25/40	20.4%
nominal	$R_{1,\sigma=5}$	5	38/40	9.4%
nominal	PD	10	29/40	$\leq$ 17%
nominal	$R_{1,\sigma=5}$	10	—	$\leq$ 6%
+50% mass	PD	10	0/40	42–49%
+50% mass	$R_{1,\sigma=5}$	10	32/40	5–7%

Pr[collision | fallback] = 0/40 for every method. CBF-disabled: both methods 0/40 (40/40 collisions).

5.4 Multiple-Comparison Correction

Table 12: Holm–Bonferroni for $m = 13$ primary family (paired McNemar $R_{1,\sigma=5}$ vs. PD on NARROW), family-wise $\alpha = 0.05$.

Table 13: Holm–Bonferroni ($m = 13$ primary family).

Rank	Test	p (raw)	Reject ($m = 13$)?
1	crossconfig +50% mass K=10	9.3×10^{-10}	Yes
2	negabl A_3-vs-PD K=5	2.8×10^{-9}	Yes
3	main narrow K=10	2.2×10^{-7}	Yes
4	main narrow K=5	8.4×10^{-6}	Yes
5	crossconfig +50% drag K=10	2.4×10^{-4}	Yes
6	main narrow K=20	4.0×10^{-5}	Yes
7	crossconfig nominal K=10	1.2×10^{-4}	Yes
8–13	cross-task ΔTErr (2×), A_3-vs-R_1 (3×)	≥ 0.04	No

7/13 reject at family-corrected level. Extending to $m = 28$ yields the same qualitative conclusions.